Forward, PINNStudio best
Inverse, PINNStudio best
DeepXDE's settings / the paper's method
Reported in the paper
ODEs
Lotka-Volterra ODE
Lorenz ODE – unknown C_1, C_2, C_3
Integro-differential ODE
1D, steady
Euler beam 1D
Poisson 1D – Dirichlet/Periodic
Poisson 1D – unknown forcing field
Fractional Poisson 1D – unknown order α
1D, time-dependent
Burgers 1D – Raissi et al.
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Schrödinger 1D
Klein-Gordon 1D
Wave 1D – space-time multi-scale Fourier features
Diffusion-reaction 1D
Black-Scholes 1D – European call
2D, steady
Helmholtz 2D – Dirichlet
Linear elasticity 2D – plate
Navier-Stokes 2D – Kovasznay flow
Poisson 2D – disk
2D, time-dependent
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Burgers 2D – Mathias et al.
Heat 2D
3D
Allen-Cahn 3D – Wight & Zhao
Navier-Stokes 3D – Beltrami flow
d, a
C_1, C_3, C_2
EI
α
λ_1, λ_2
γ_2, γ_1
c
γ
C
D
k_0
μ, λ
Re
f
c_2, c_1^2
ε^2
ν
α
$\lambda\varepsilon^2$, λ
ν
10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0}
Relative L_2 error of the main output at the last reference time
0.001 0.1 10
Parameter error (%)